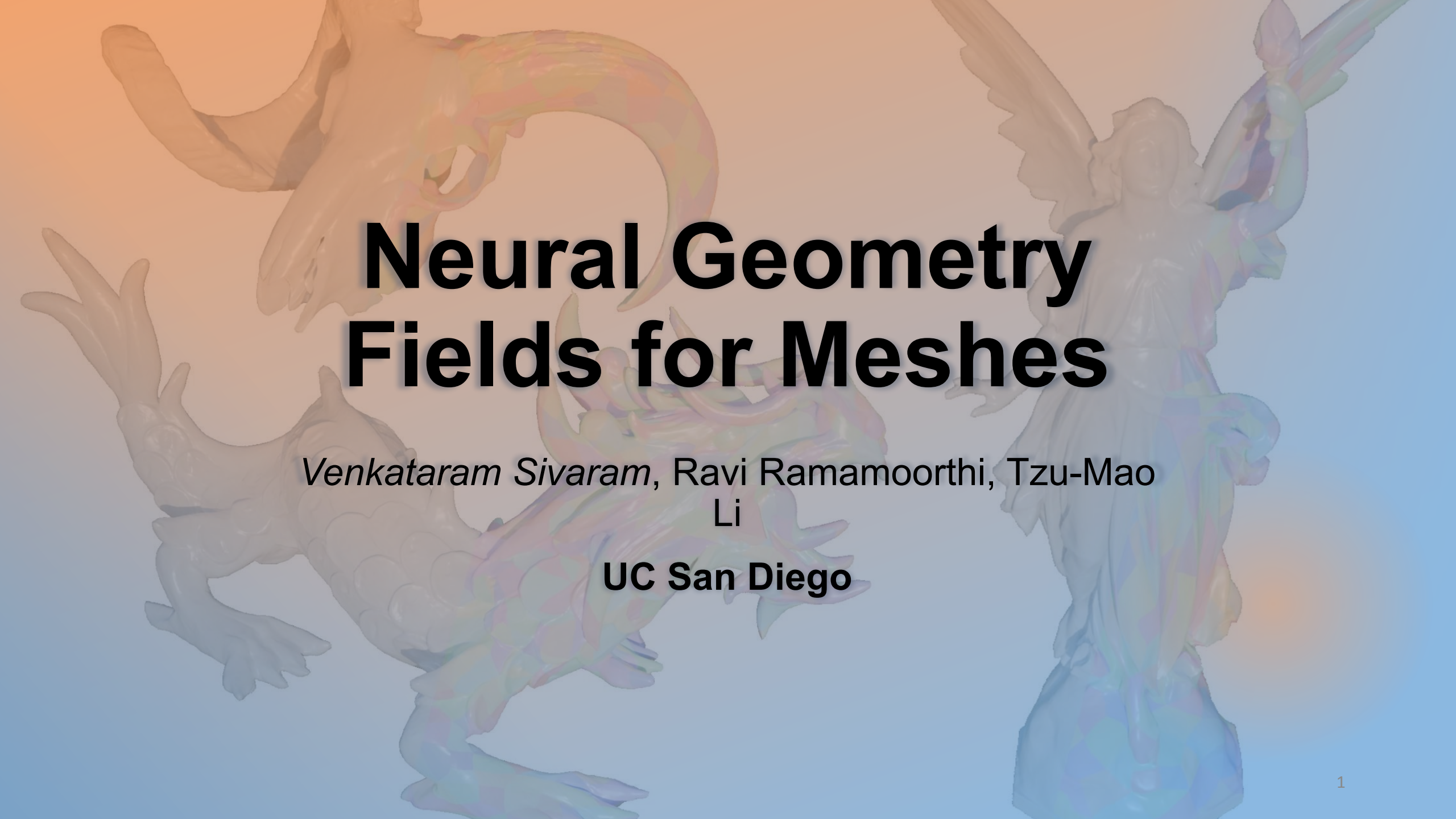


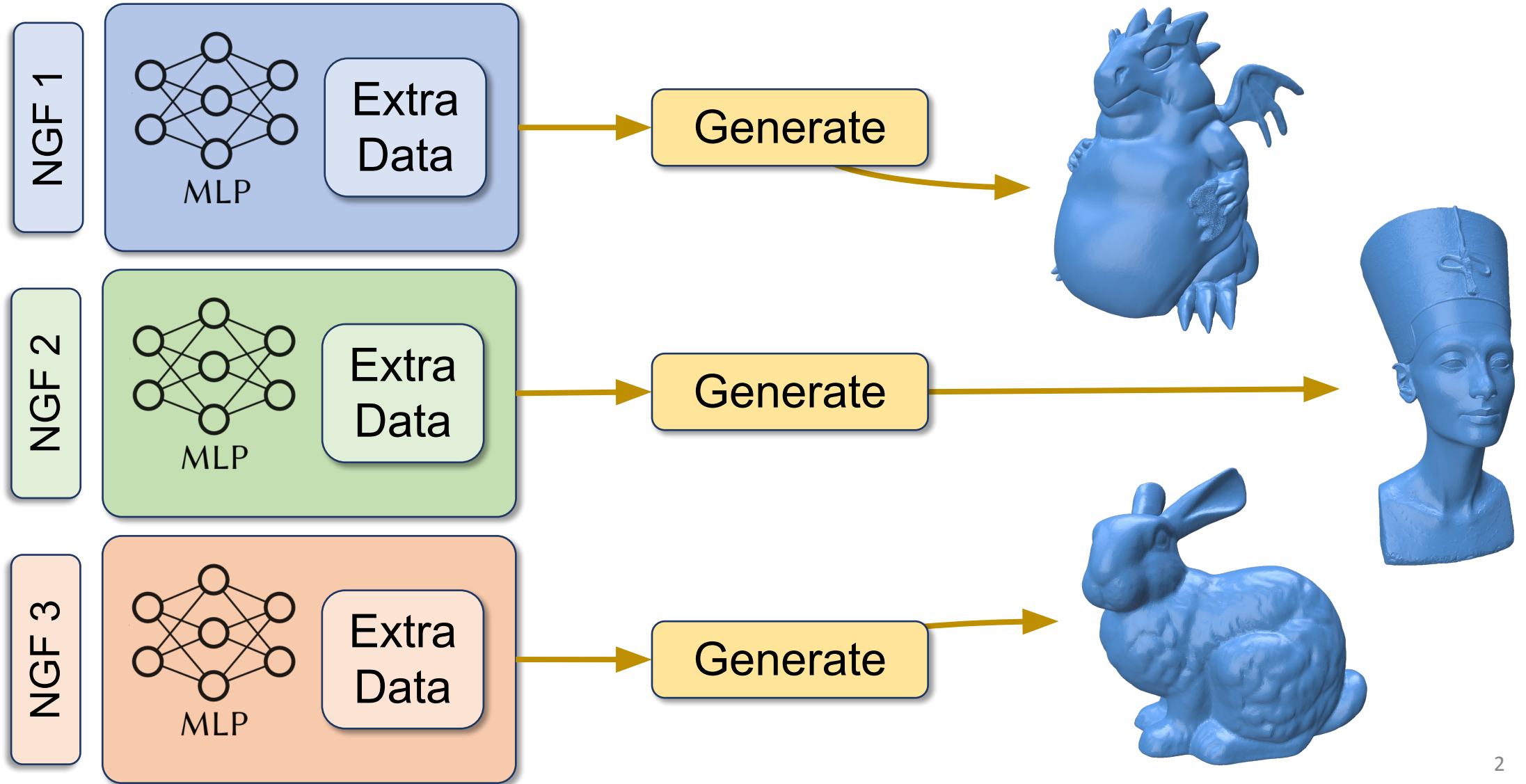


Neural Geometry Fields for Meshes

Venkataram Sivaram, Ravi Ramamoorthi, Tzu-Mao
Li

UC San Diego

Goal: represent meshes with NNs



Previous work in neural surfaces

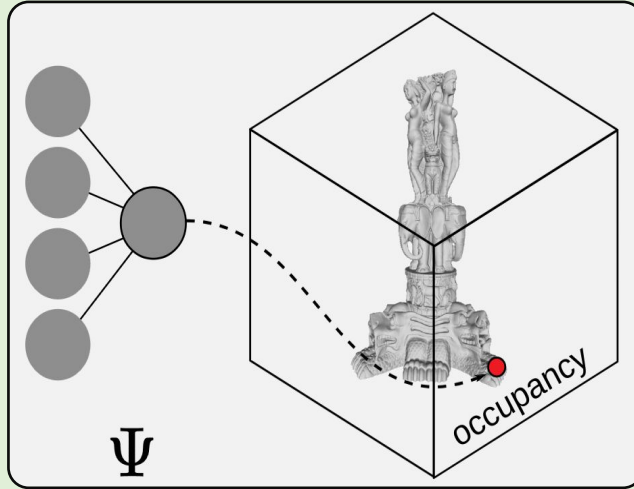
Occupancy

Occupancy Networks
[Mescheder2018]



Distance fields

DeepSDF, ACORN
[Park2019, Martel2021]



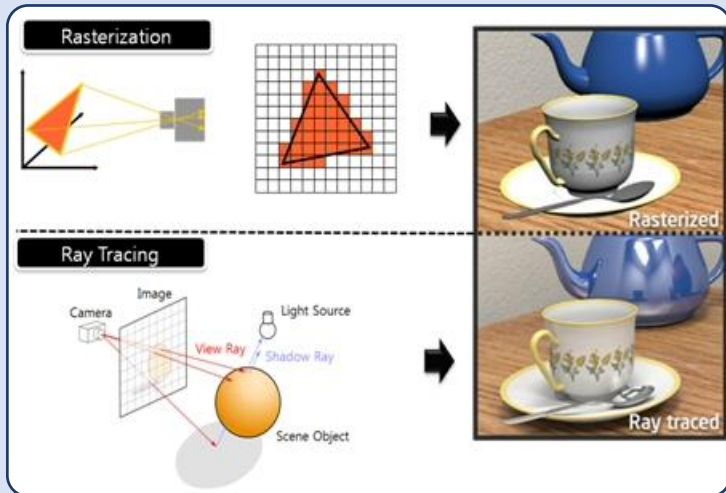
Density fields

NeRF, 3DGS
[Mildenhall2020, Kerbl2023]

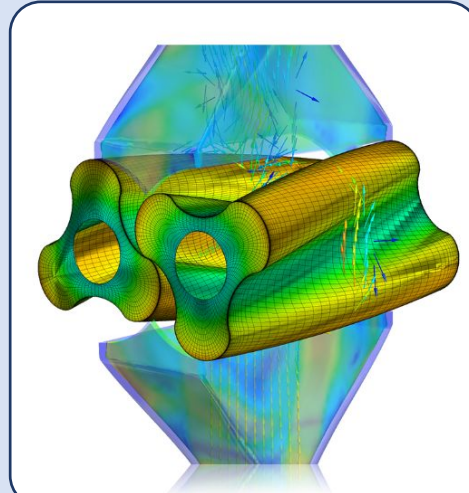


Triangle meshes are ubiquitous

Rendering



Physics Simulation

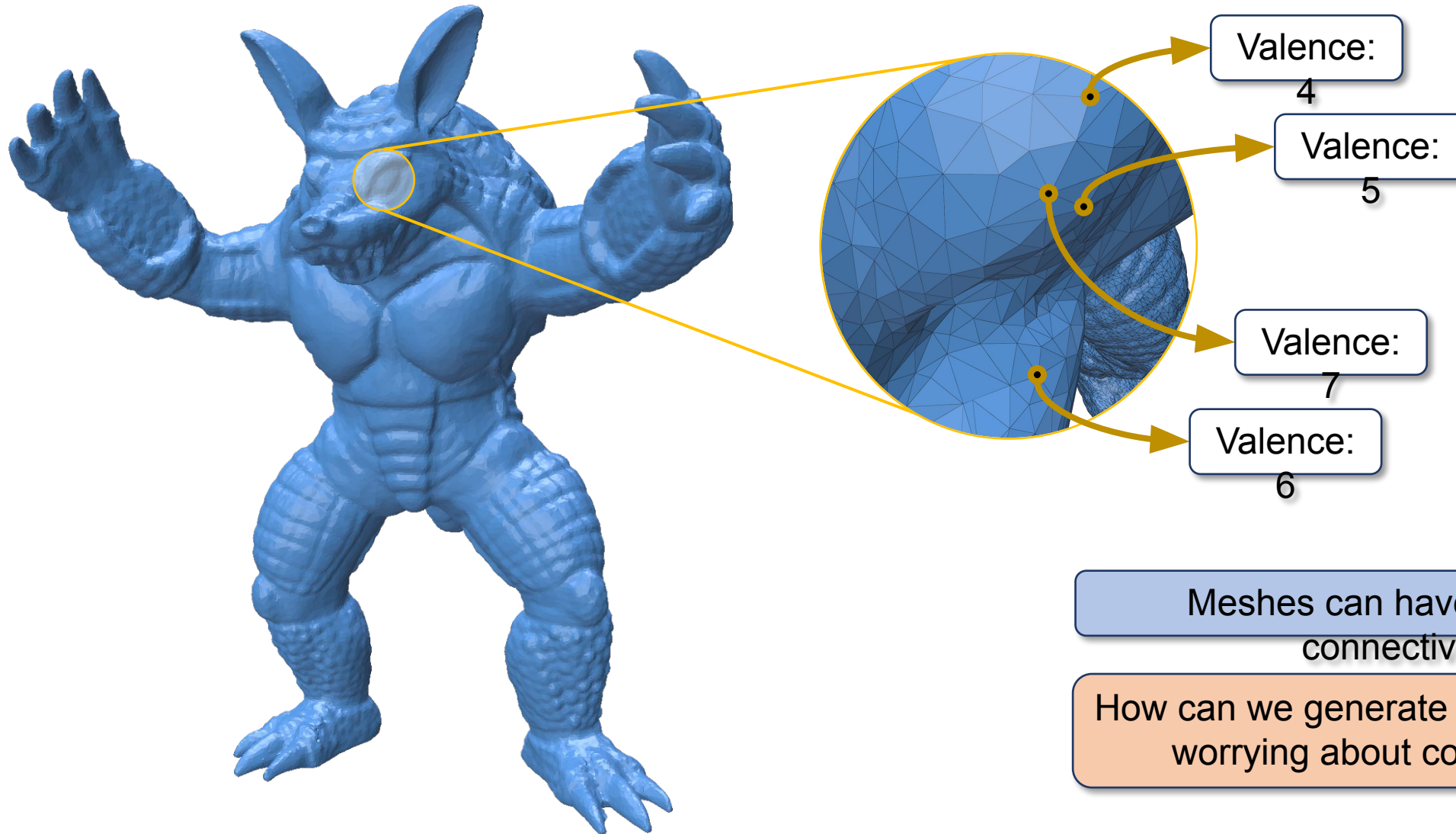


Sculpting and Modeling



Previous methods need to be converted to meshes during preprocessing.

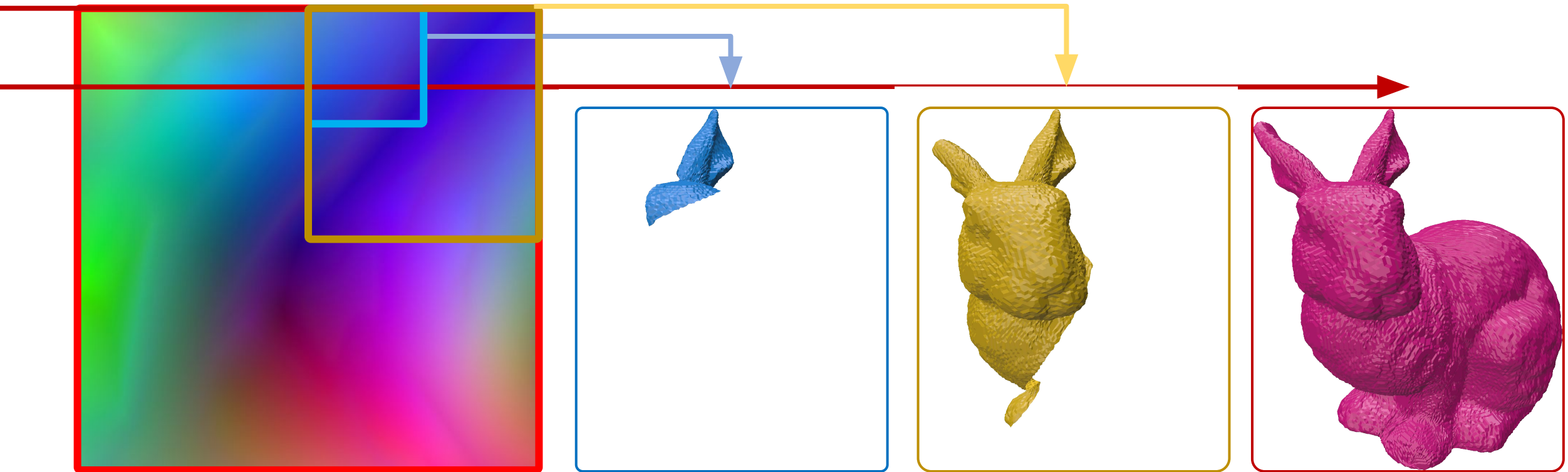
Challenges with meshes in the wild



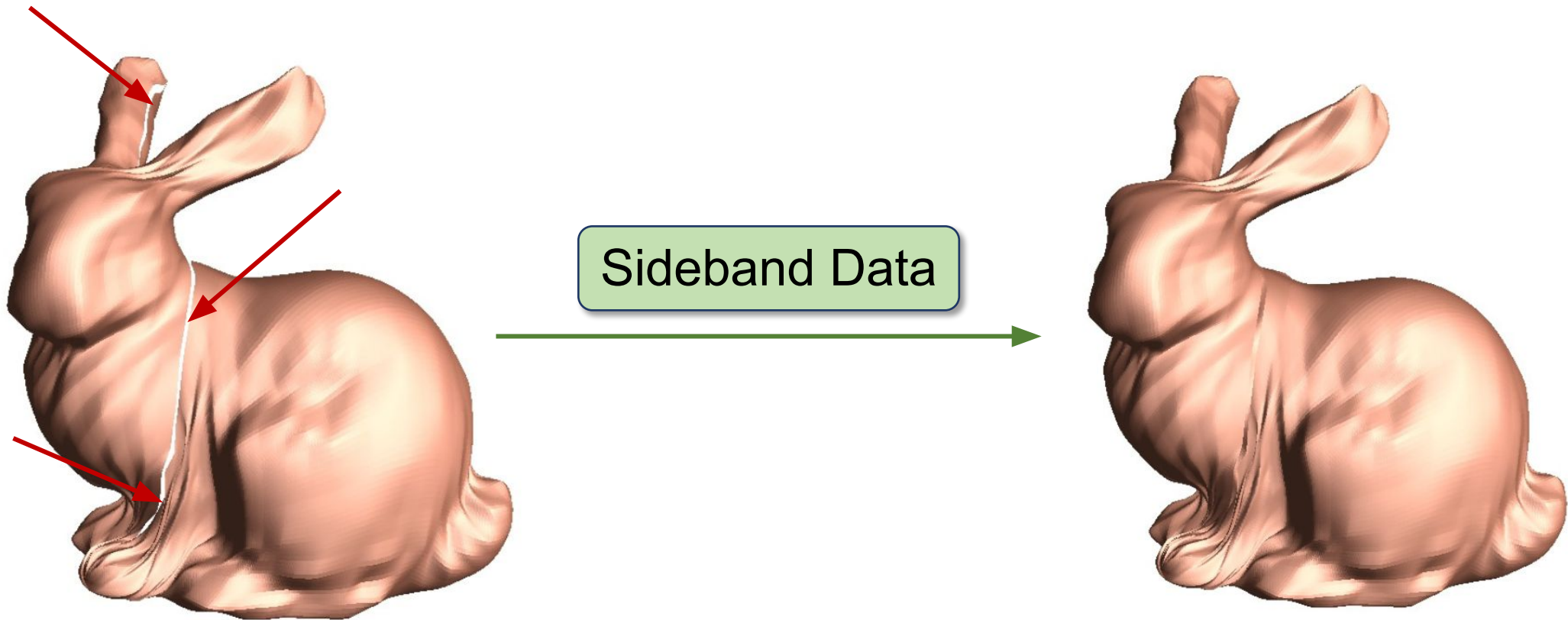
Meshes can have irregular connectivity!

How can we generate vertices without worrying about connectivity?

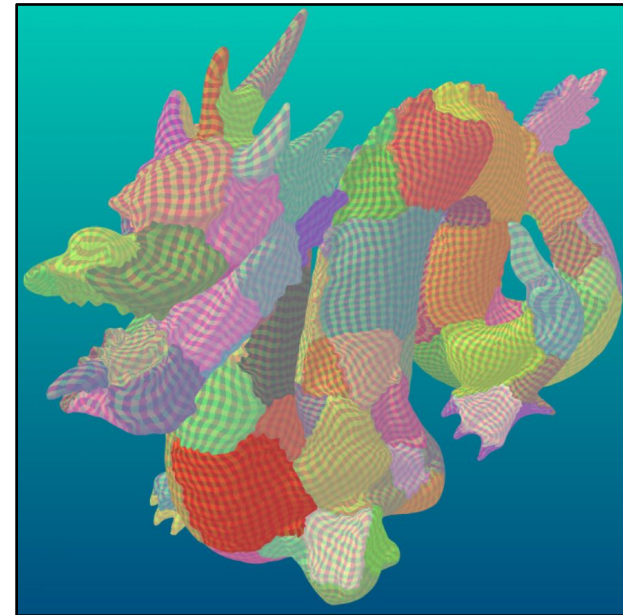
Geometry images: a uniform layout



Downsides to geometry images



Multichart geometry images



Our idea: neural geometry fields

IDEA 

Start with rectangular geometry images.

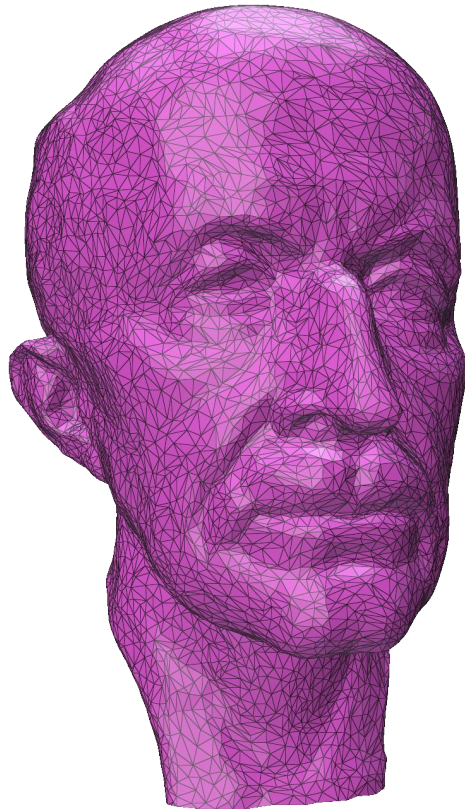
- ◆ How do we start with rectangular geometry images?
- ◆ How do we improve the quality of these geometry images?

Generating a base surface

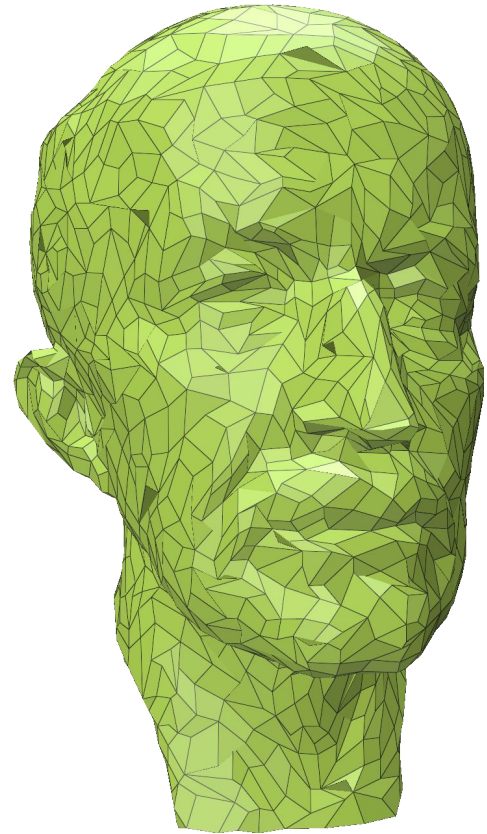
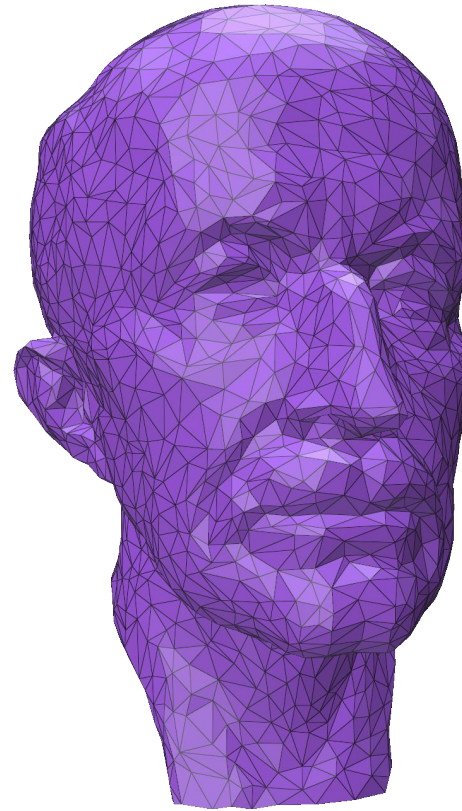
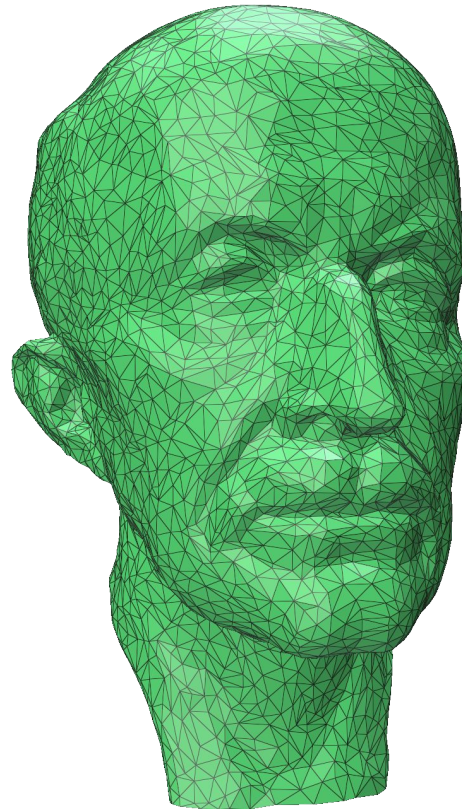
Original



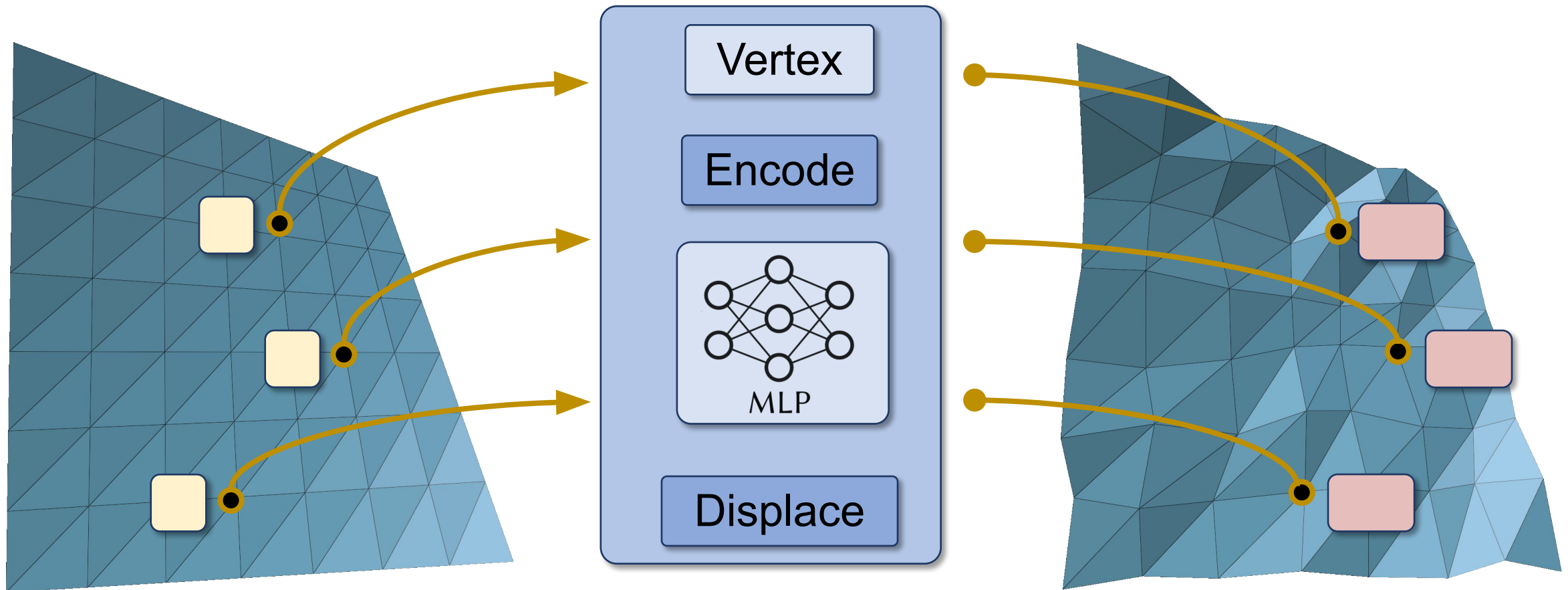
Mesh simplification



Quads

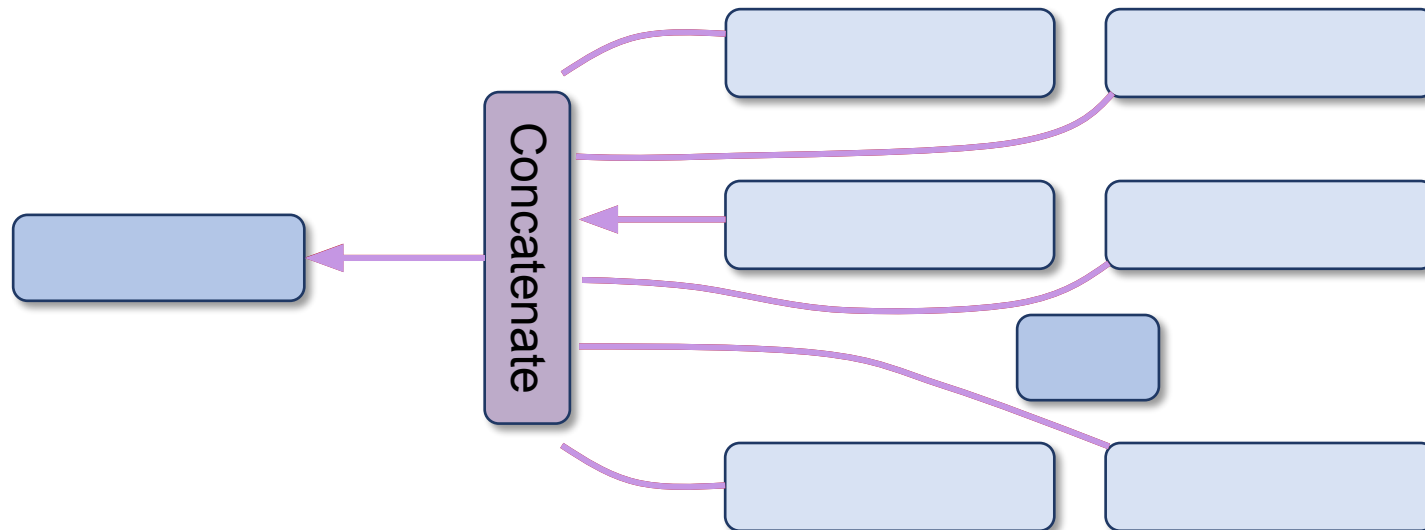


Displacements with geometry fields

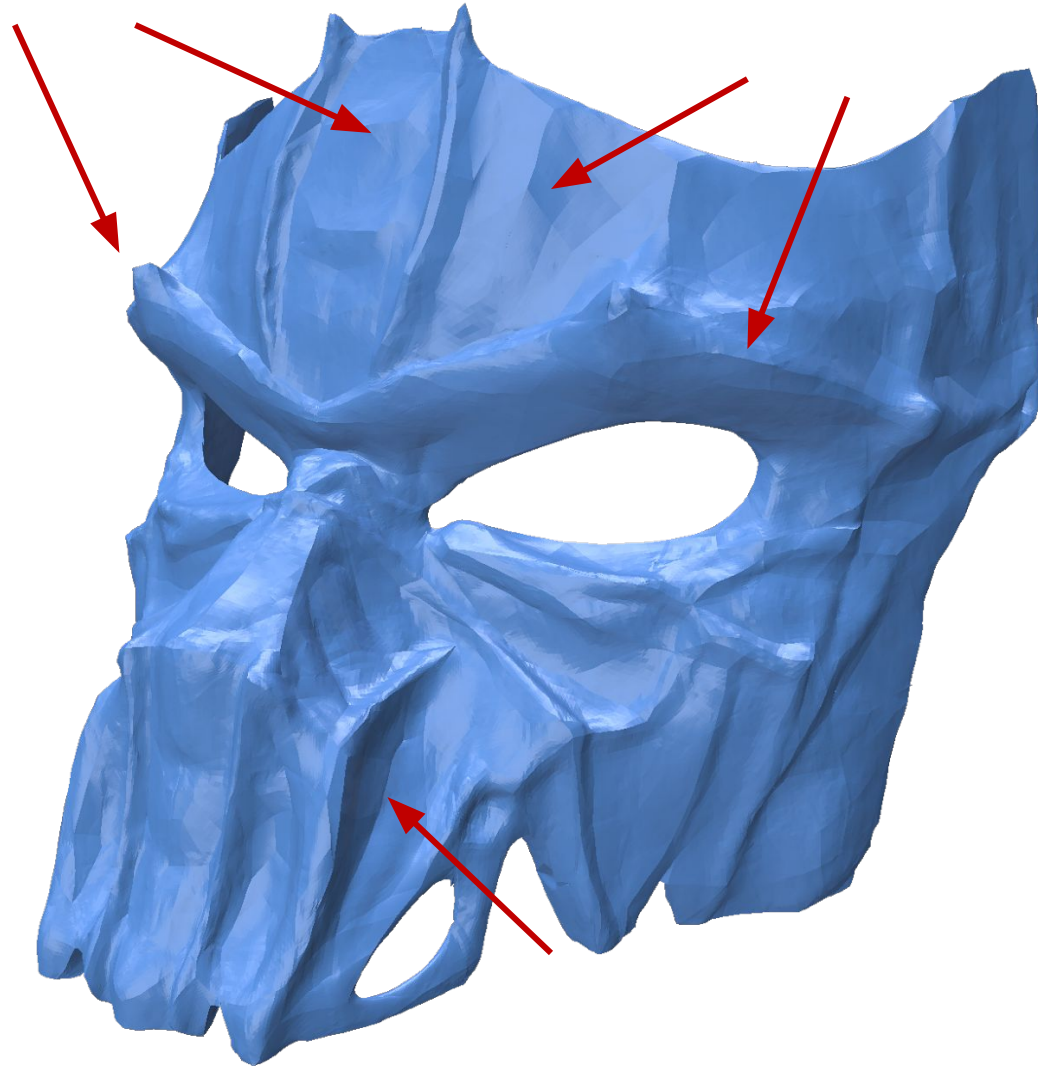


Inputs to geometry fields

What should be the result of **encode**?



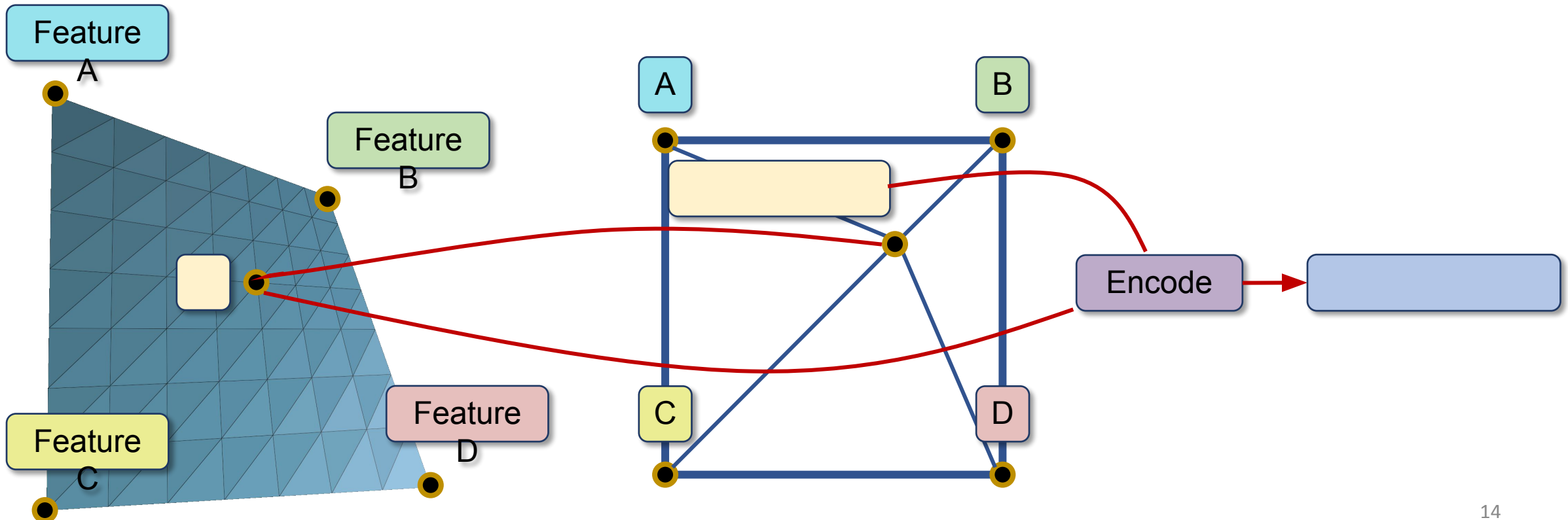
Inputs to geometry fields



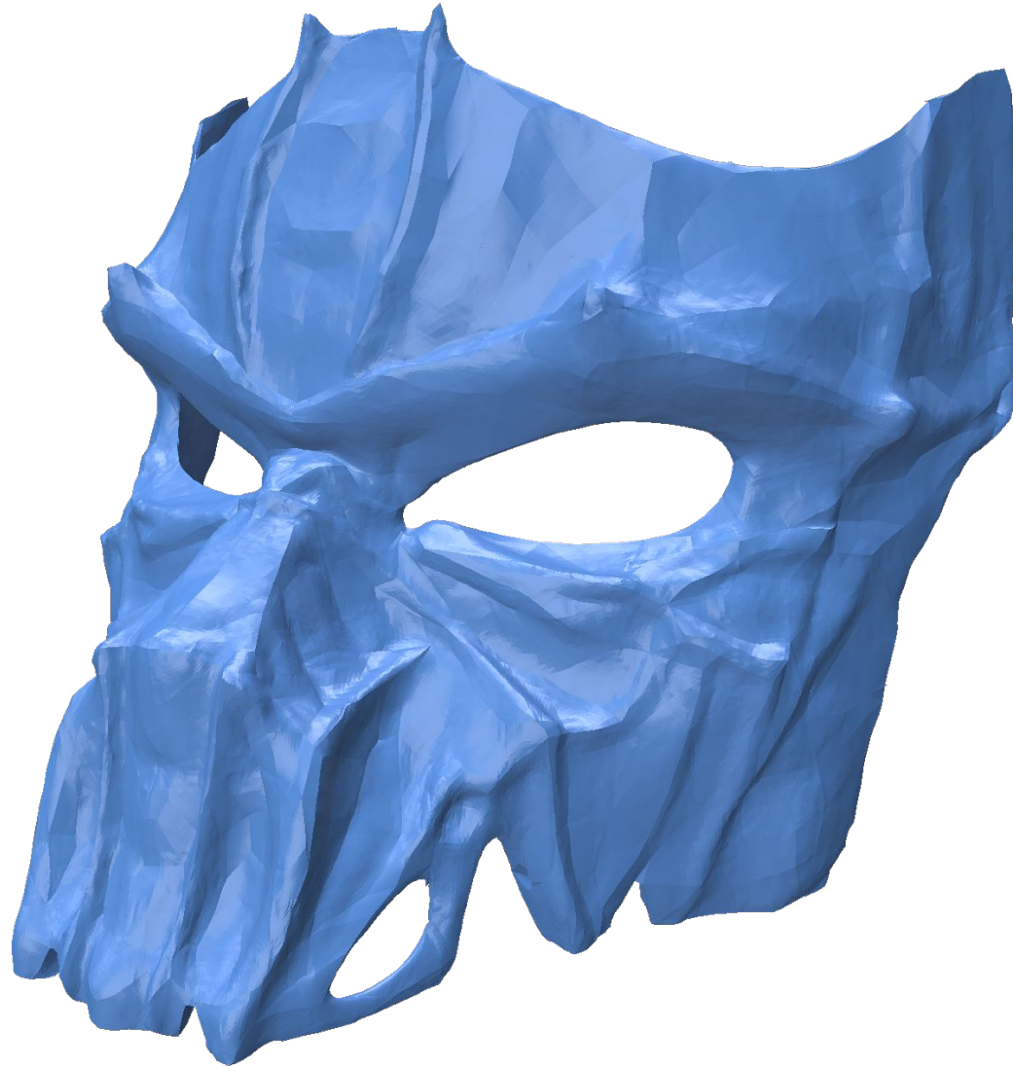
Inputs to geometry fields

IDEA 

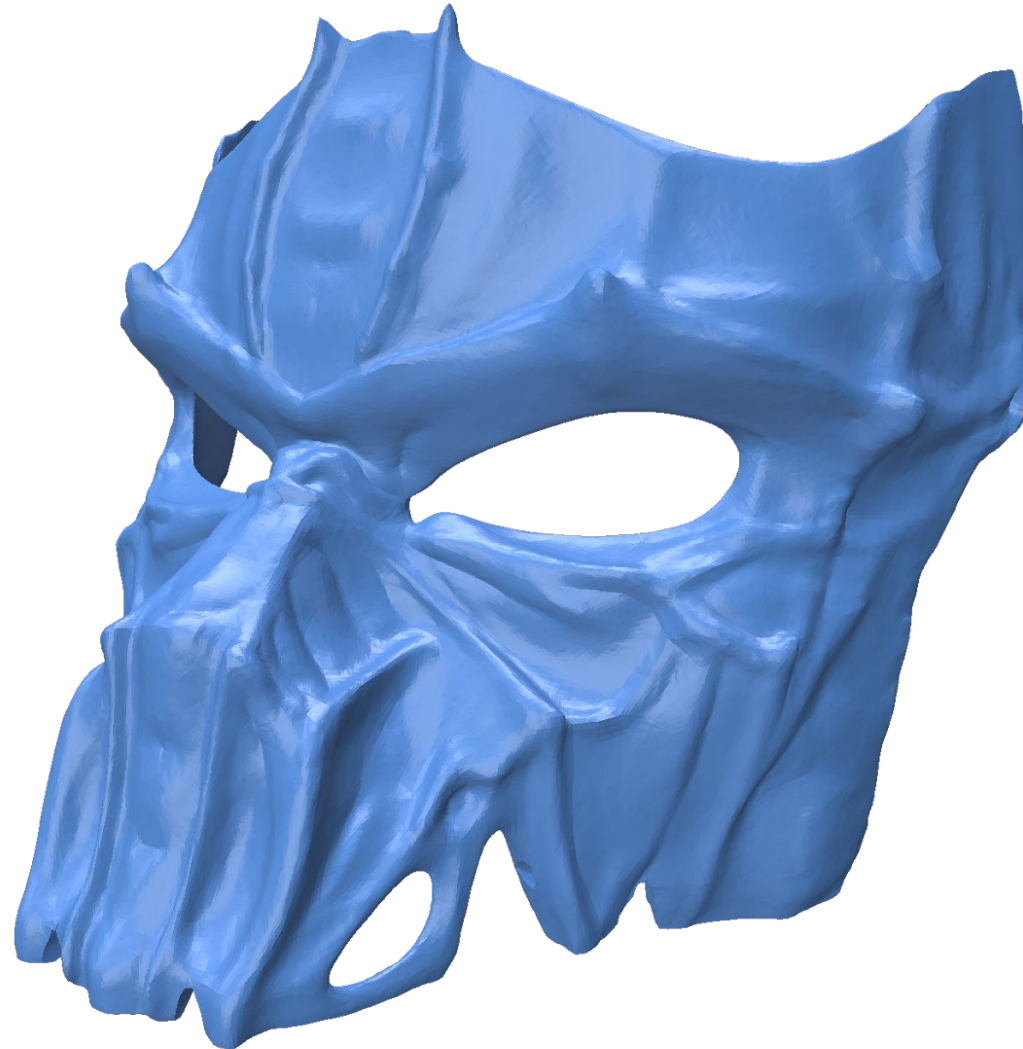
*Exploit the manifold structure of surface;
embed a **feature field** within the surface.*



Inputs to geometry fields

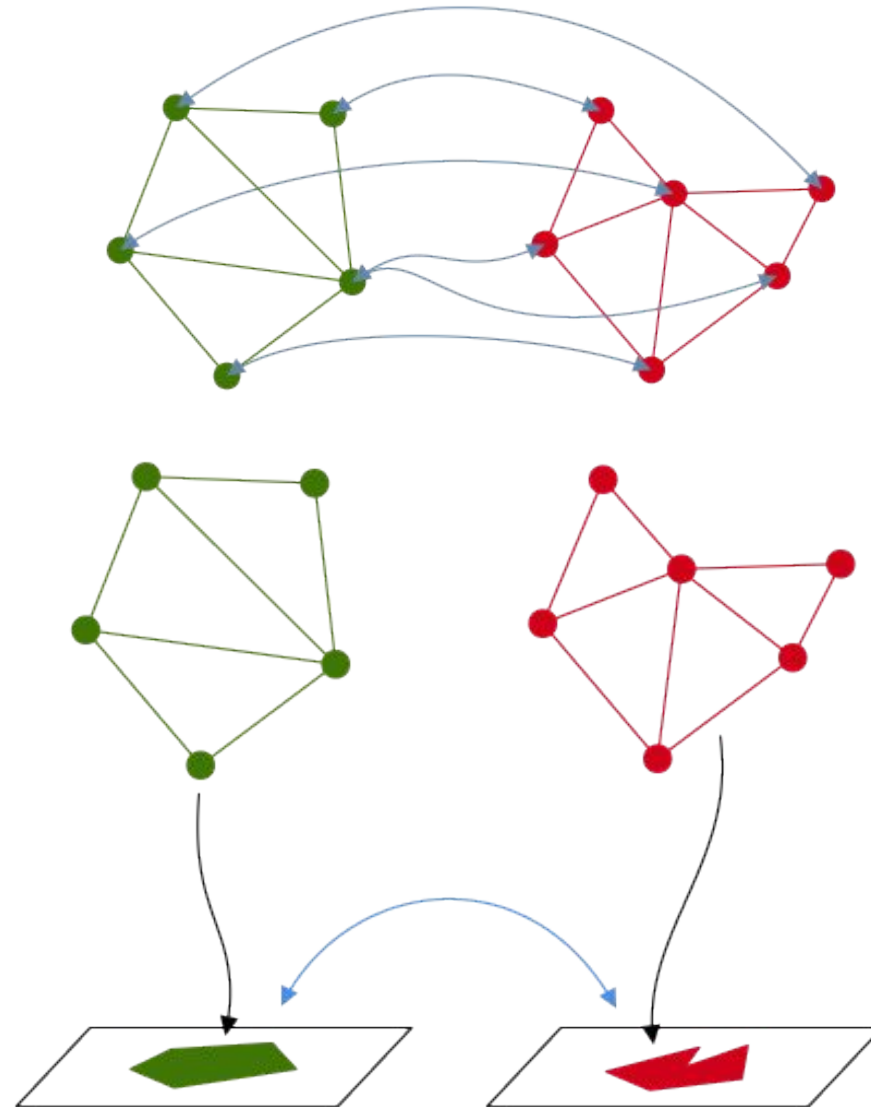


Inputs to geometry fields

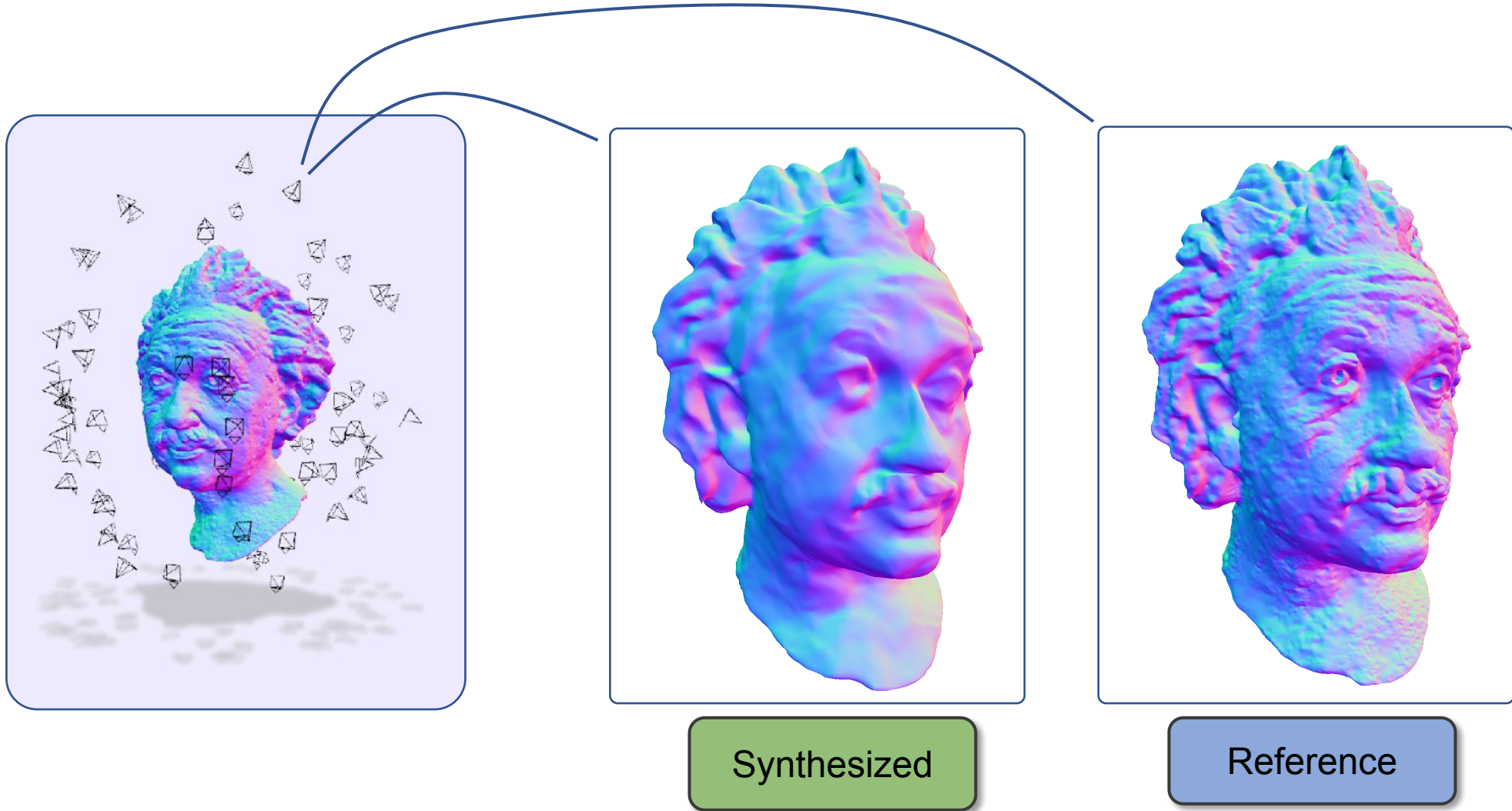


Suitable loss functions

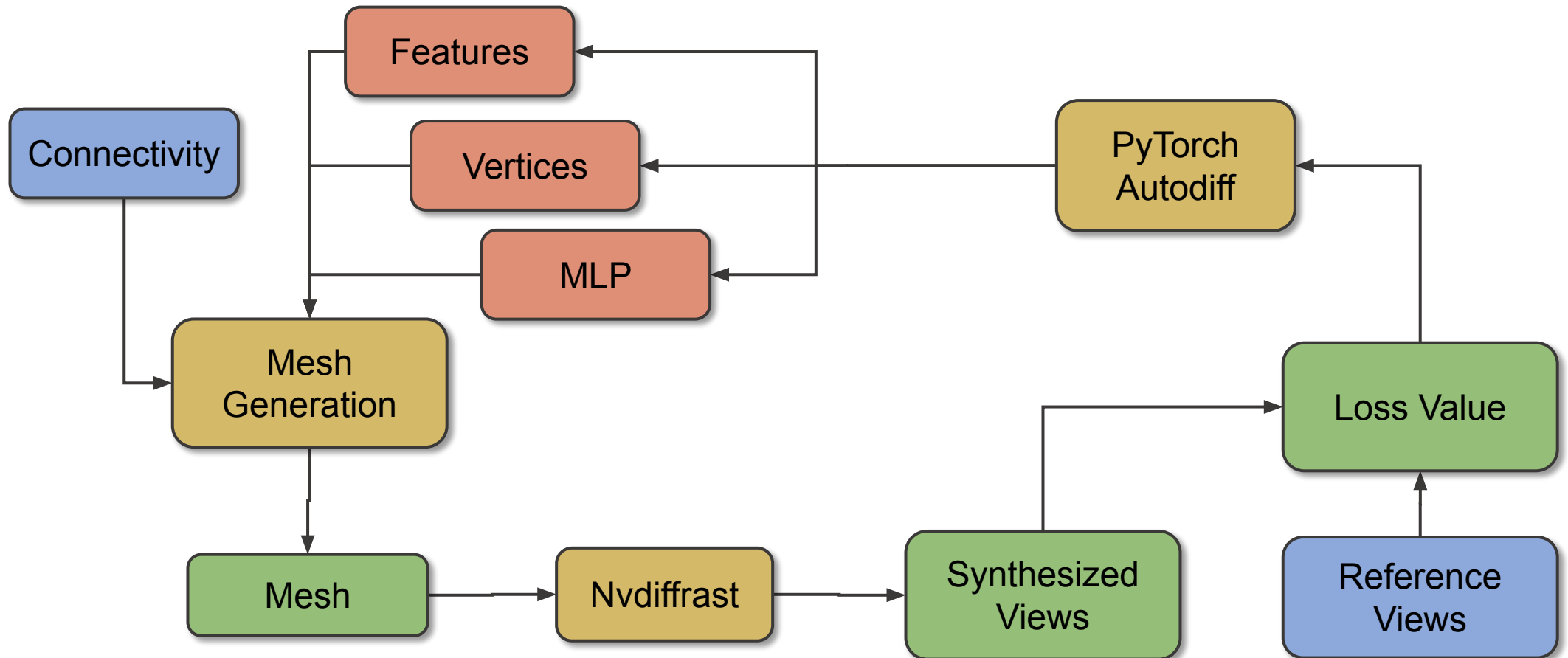
*Vertex-based
correspondence*
(i.e. Chamfer distance)



Inverse rendering for surface recovery

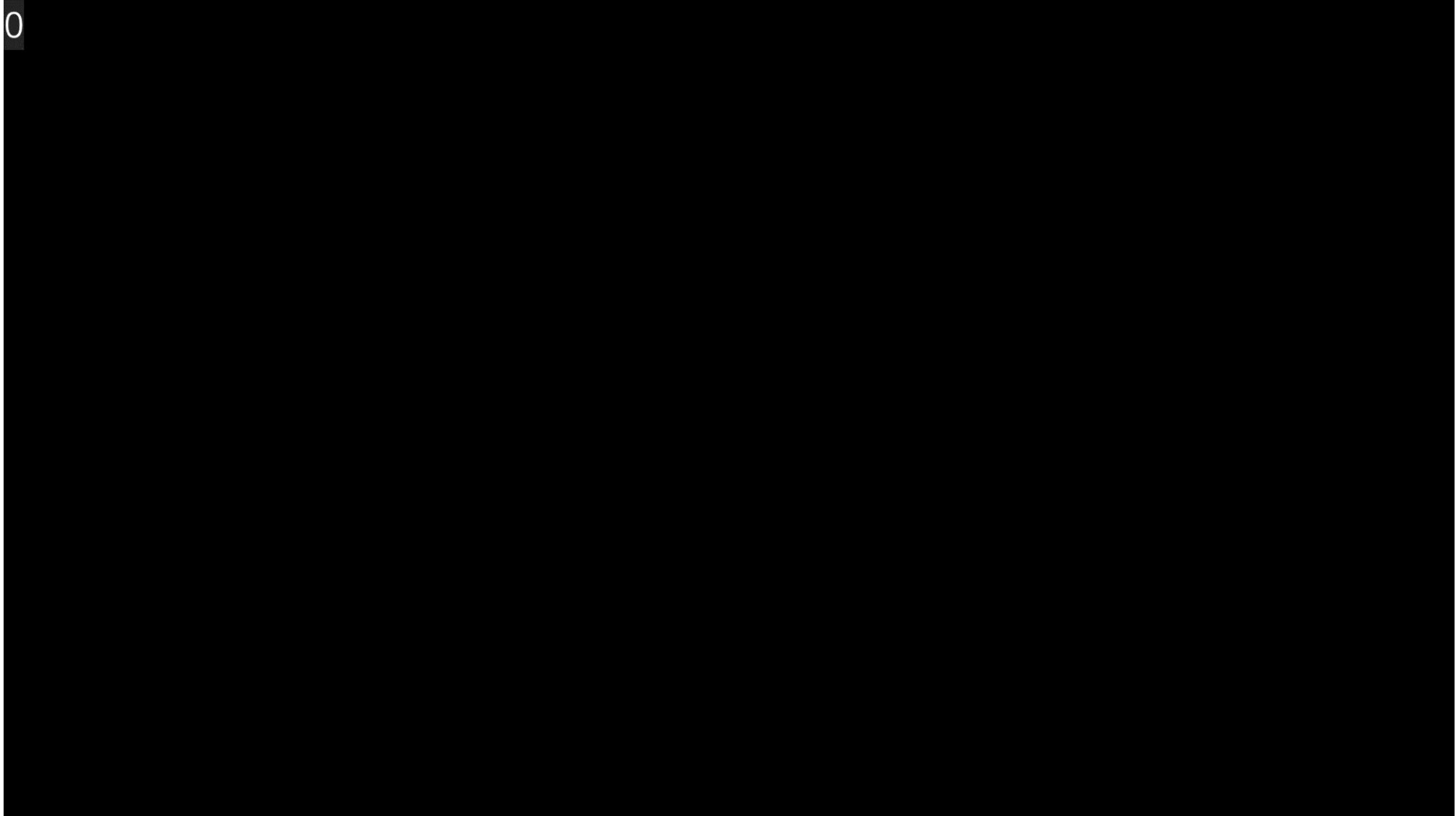


Optimization pipeline

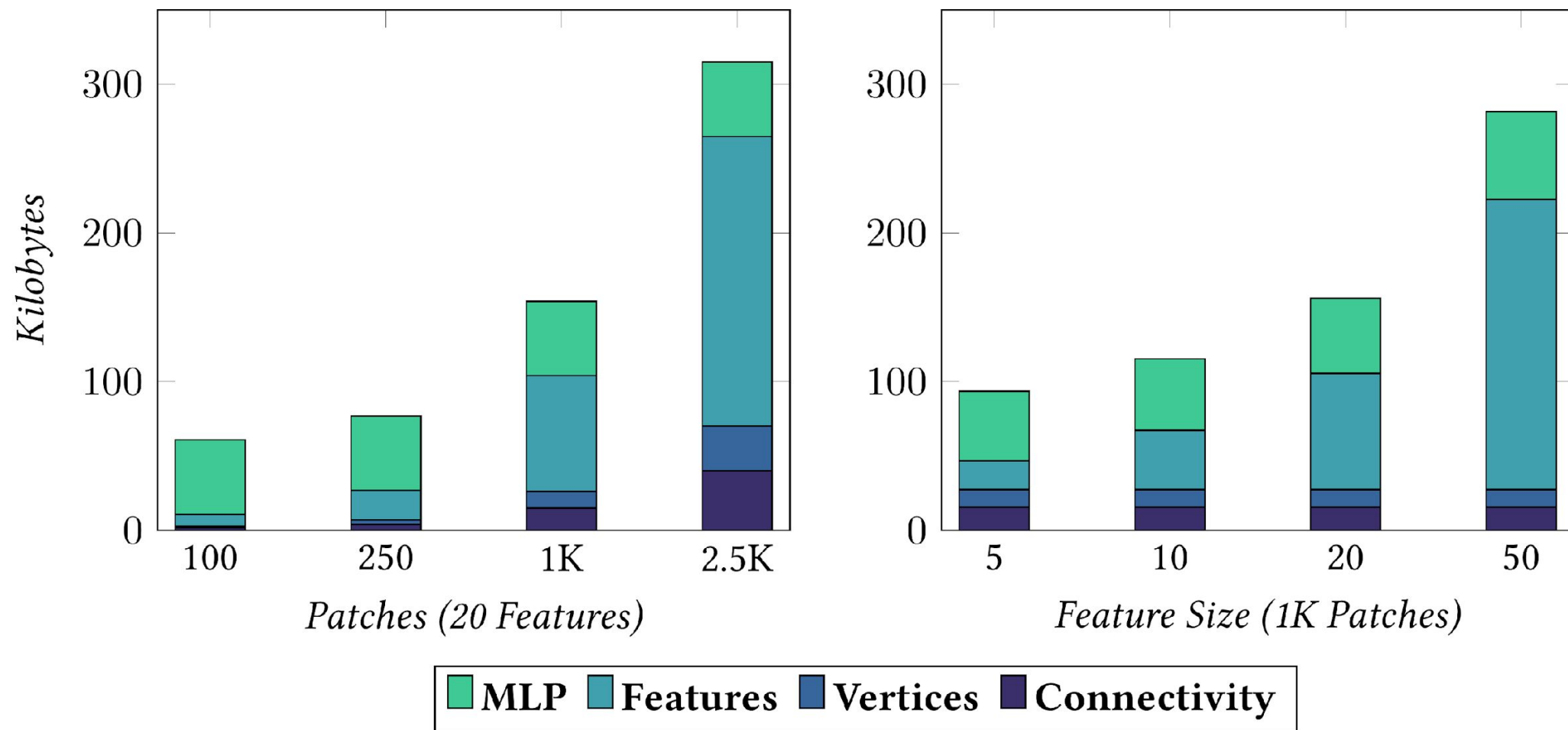


Fast optimization progress

0



Storage breakdown of NGFs



Baseline methods

- ◆ *Qslim*

Simplification via surface intrinsic metrics.

- ◆ *Nvdiff(modeling)*

Inverse rendering to optimize Qslim-ed surfaces.

- ◆ *Draco*

Quantize vertex information.

Equal size comparison



Qslim (362 KB)



Nvdiff (362 KB)



NGF (362 KB)



Target (41 MB)

Equal size comparison



Qslim (362 KB)



Nvdiff (362 KB)



NGF (362 KB)



Target (15 MB)

Comparison versus Draco



Draco-8-bit
(393 KB)



Draco-10-bit
t



NGF
(156 KB)



Target
(18 MB)

Real-time rasterization with meshlets



Summary of NGFs

NGF is a *hybrid neural architecture* for directly representing meshes.



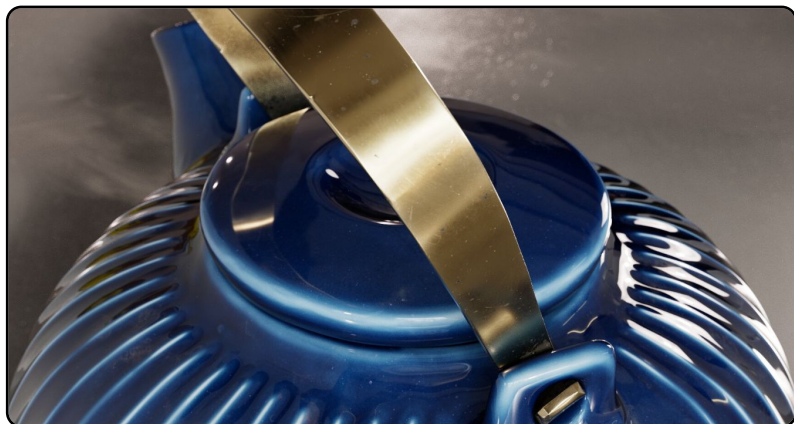
Key Ideas



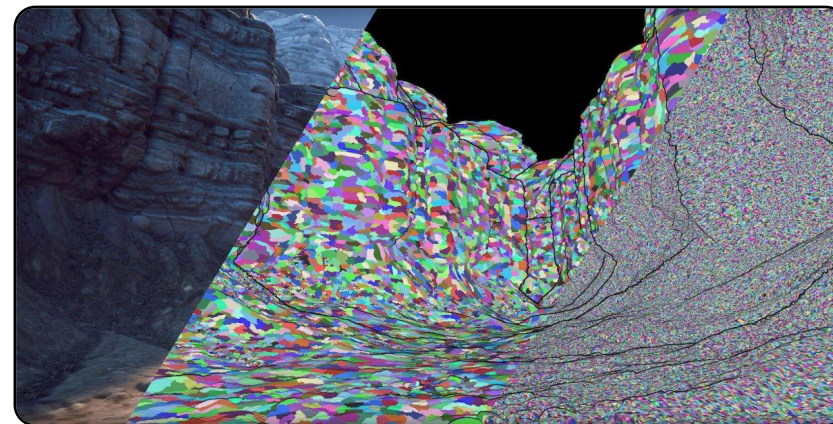
- ◆ *MLP input space is a 2-manifold rather than the ambient space.*
- ◆ *Multiview inverse rendering is smooth and effective.*

Future directions with NGFs

◆ *Unify geometry and materials*



◆ *Dynamic level-of-detail*



◆ *Compressed 3D surface scans*



Thanks!